

Emin Yusuf AYDIN

POSTDOC · BIOENGINEERING PROGRAM

Hacettepe University, Beytepe Campus, Ankara, Türkiye 06800

✉ eminyusufaydin@gmail.com | 📧 eminyusuff | 🌐 eminyusufaydin

Research Interests

I investigate how animals actively shape sensory information and integrate multiple sensory modalities to generate adaptive behavior. My research combines closed-loop behavioral experiments in weakly electric fish with system identification, computational modeling, and feedback-control approaches to characterize the dynamical principles underlying active sensing and sensorimotor coordination. I am also interested in neurocomputing, neuromorphic systems, and brain-inspired robotics, particularly in translating biological principles of sensing, learning, and adaptive control into intelligent systems capable of robust operation in uncertain and changing environments.

Education

Hacettepe University

PH.D. IN BIOENGINEERING

Ankara, Türkiye

Oct 2020 – 27.07.2026

- Advisor: Assoc. Prof. Dr. İsmail Uyanık
- Dissertation: Modeling the Role of Active Sensing Movements During Refuge Tracking Behavior in Weakly Electric Fish
- **TÜBİTAK 1002-A Short-Term Support Module – Short-Term Adaptive Modulation of Refuge-Tracking Behavior Under Controlled Sensory Perturbations:** Principal Investigator; accepted for funding in 2026. The project investigates short-term sensorimotor adaptation in weakly electric fish using software-controlled sensory perturbations during closed-loop refuge tracking.
- **TÜBİTAK 1001 Project No. 123E155 – System Identification of Multisensory Integration and Dynamic Sensory Weighting:** Doctoral researcher; designed and conducted multisensory refuge-tracking experiments under visual, refuge-geometry, and water-flow manipulations; synchronized camera-based motion tracking with actuator control and data acquisition; and applied DFT/FFT, cross-spectral analysis, transfer-function estimation, system identification, computational modeling, bioinspired control, and scientific visualization using MATLAB and Python.
- **TÜBİTAK 1001 Project No. 120E198 – Behavioral Mechanisms of Active Sensing:** Full-time doctoral project fellow; developed and operated the electronic and experimental infrastructure for active-sensing studies, including camera-based tracking, embedded controllers, data-acquisition hardware, and ROS/ROS2-based real-time control; and performed data preprocessing, signal processing, behavioral time-series analysis, statistical modeling, motion tracking, machine-learning-assisted landmark detection, adaptive FIR filtering, closed-loop feedback manipulation, and real-time experimental software development using MATLAB, Python, ROS/ROS2, Arduino, Raspberry Pi, and STM32.
- GPA: 3.97/4.00

Ankara Yıldırım Beyazıt University

M.SC. IN MUSCULOSKELETAL SYSTEM AND REGENERATIVE MEDICINE

Ankara, Türkiye

Jan 2017 – Jul 2020

- Advisor: Prof. Dr. Murat Bozkurt
- Thesis: Comparison of the Combined and Individual Effects of Stromal Vascular Fraction and Bone Marrow Concentrate on the Achilles Tendon
- **TÜBİTAK Project No. 119S901 – Combined and Individual Effects of Stromal Vascular Fraction and Bone Marrow Concentrate on Achilles Tendon Healing:** Graduate project fellow; contributed to the in vivo injury model, evaluation of regenerative treatments, outcome assessment, statistical data analysis, interpretation of findings, and manuscript preparation; and developed experience in preclinical experimental design, laboratory-animal procedures, musculoskeletal outcome assessment, quantitative data analysis, and scientific reporting.
- GPA: 3.91/4.00

Başkent University

B.SC. IN PHYSIOTHERAPY AND REHABILITATION

Ankara, Türkiye

Aug 2009 – Jun 2013

- GPA: 3.23/4.00

Publications

JOURNALS

- Aydın, E. Y.**, Unlutabak, B. & Uyanık, I. (2025). Flow impairs multisensory tracking and increases active sensing in weakly electric fish. *Journal of Experimental Biology* (WOS-Q1), jeb-251478. doi:10.1242/jeb.251478
- Karagoz, O. K., Kilic, A., **Aydın, E. Y.**, Ankarali, M. M., & Uyanık, I. (2025). Predictive Uncertainty in State-Estimation Drives Active Sensing in Weakly Electric Fish. *Bioinspir. Biomim.* (WOS-Q2) 20 016018,.doi:10.1088/1748-3190/ad9534.
- Aydın, E. Y.**, Aşık, M., Aydın, H. M., Çay, N., Gümüşkaya, B., Çağlayan, A., Torabi, A., Yüksel, S., Veizi, E., & Bozkurt, M. (2023). The co-use of stromal vascular fraction and bone marrow concentrate for tendon healing. *Current stem cell research & therapy*, 18(8), 1150-1159. doi: 10.2174/1574888X18666230221141743.
- Yüksel, S., Aşık, M. D., Aydın, H. M., Tönük, E., **Aydın, E. Y.**, & Bozkurt, M. (2022). Fabrication of a multi-layered decellularized amniotic membranes as tissue engineering constructs. *Tissue and Cell* (WOS-Q2), 74, 101693. doi:10.1016/j.tice.2021.101693.
- Seyhan, R. G., Tunay, V. B., Göksülük, D., **Aydın, E. Y.**, & Ergun, N. (2020). Bedensel engelli yüzücülerde çıkış süresinin gövde esnekliği, aerobik endurans ve anaerobik güç ile ilişkisi. *Journal of Exercise Therapy and Rehabilitation*, 7(2), 186-192.

BOOK CHAPTER

- Aydın, E. Y.**, & Uyanık, I. (2023). A Case Study as a Multisensory Integartion Model: Weakly Electric Fish. Editor:Dinçer PR, Use of Zebrafish as a Model Organism. 1. Edition. Ankara: Türkiye Klinikleri., p.104-9.

CONFERENCE PROCEEDINGS

- Aydın, E. Y.**, Yılmaz, O., Öztürk, M., Doğan, H., & Uyanık, İ. (2026, August 23–28). Real-Time Decomposition of Active Sensing and Tracking Motions via a Structure-Aware Filtering Framework. 23rd IFAC World Congress (IFAC WC 2026), Busan, South Korea (accepted).
- Aydın, E. Y.**, Yılmaz, O., Öztürk, M., & Uyanık, İ. (2026, July 7–10). Structure-Aware Filtering for Real-Time Separation of Active-Sensing and Tracking Motions. 34th Signal Processing and Communications Applications Conference (SIU), İstanbul, Türkiye (accepted).
- Topçakar, S., **Aydın, E. Y.**, Demirel, A., & Uyanık, İ. (2024, 15-18 May). Reshaping Active Sensing via Closing a Feedback Loop Around Free Behavior. 32nd Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE.
- Çatalbaş, B., Elikuru, D., **Aydın, E. Y.**, & Uyanık, İ. (2023, 5-8 July). Tracking the Nodal Point of Weakly Electric Fish Using Artificial Neural Networks. 31st Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE.
- Koç, O., Demirel, A., **Aydın, E. Y.**, İbişoğlu, F., Solmaz, Ş. İ., Arı, K., Idman, A., & Uyanık, İ. (2022, 15-18 May). System Identification of the Target Tracking Behavior of Zebrafish During Rheotaxis. 30th Signal Processing and Communications Applications Conference (SIU) (pp. 1-4). IEEE.

CONFERENCE AND SYMPOSIUM PRESENTATIONS

- Aydın, E. Y.**, & Uyanık, I. (2026, 22-24 June). Active Sensing and Adaptive Feedback Control in Weakly Electric Fish. SOAR/BISSCITs Workshop 2026, Wageningen,NL (accepted)
- Aydın, E. Y.**, & Uyanık, I. (2025, 26-28 November). Active Sensing Strategies of Weakly Electric Fish Under Varying Flow Conditions. I. International Neuronal Energy Dynamics Congress, Ankara, Türkiye. *Gazi Medical Journal*, 37(Supplement 1), 5.
- Muratoğlu, M., Yılmaz, O., **Aydın, E. Y.**, Doğan, H., & Uyanık, I. (2025, 26-28 November). Low-Latency Multi-Notch Filtering for Active Sensing in Weakly Electric Fish. I. International Neuronal Energy Dynamics Congress, Ankara, Türkiye. *Gazi Medical Journal*, 37(Supplement 1), 8.
- Aydın, E. Y.**, & Uyanık, I. (2025, 15-19 November). Adaptation of Active Sensing Responses to Sensory Manipulations in Weakly Electric Fish. SFN Meeting 2025 (NEUROSCIENCE 2025).San Diego,CA,USA.
- Aydın, E. Y.**, Karagoz, O. K., Ankarali, M. M., Demirel, A., & Uyanık, I. (2024, 5-9 October). A fish-in-the-loop comparison of various active sensing models. SFN Meeting 2024 (NEUROSCIENCE 2024).Chicago,IL,USA.
- Dipi, F. S., Muratoğlu, M., **Aydın, E. Y.**, & Uyanık, I. (2024, 8-11 September). Closed-loop modulation of active sensing reveals the critical role of sensory feedback. 10th IFAC Conference on Foundations of Systems Biology in Engineering (FOSBE 2024), Corfu Island, Greece.

- Aydin, E. Y.,** & Uyanik, I. (2024, 25-29 June). A Fish-in-the-loop System to Study the Underlying Mechanisms of Active Sensing, FENS FORUM 2024, Vienna, Austria.
- Aydin, E. Y.,** Yilmaz, O., Topcakar, S., & Uyanik, I. (2024). Closed-loop Manipulation of Active Sensing Movements of Weakly Electric Fish Integrative and Comparative Biology (Vol. 64, Issue Supplement 1, pp. S29-S29)
- Ankarali, M.M., Karagoz, O.K., Kilic, A., **Aydin, E. Y.,** & Uyanik, I. (2024). Exploring Motion-Based Active Sensing in Relation to Predictive State-Estimation Uncertainty. Integrative and Comparative Biology (Vol. 64, Issue Supplement 1, pp. S19-S20)
- Aydin, E. Y.,** & Uyanik, I. (2023, 11-15 November). Modeling the Effects of Flow Speed on Smooth-Pursuit Tracking in Active Sensing Movements of Weakly Electric Fish. SFN Meeting 2023 (NEUROSCIENCE 2023). Washington D.C., USA.
- Aydin, E.,** & Uyanik, I. (2023). Flow Speed Affects the Smooth Pursuit Tracking and Active Sensing Movements of Weakly Electric Fish. Integrative and Comparative Biology (Vol. 63, Issue Supplement 1, pp. S26-S26).
- Aydin, E. Y.,** & Uyanik, I. (2022). Investigation of flow speed on the active sensing movements of weakly electric fish. Anatomy: International Journal of Experimental & Clinical Anatomy, 16.
- Elikuru, M. D., **Aydin, E. Y.,** Gürlü, N., Çatalbaş, B., & Uyanik, I. (2022). Tracking the anal fin and nodal point of weakly electric fish using deep learning. Anatomy: International Journal of Experimental & Clinical Anatomy, 16.
- Mardin, C. Ş., İbişoğlu, F., **Aydin, E. Y.,** Öztaş, M., & Uyanik, İ. (2022). The effects of sensory salience on the refuge tracking behavior of weakly electric fish. Anatomy: International Journal of Experimental & Clinical Anatomy, 16.

DATASETS AND CODE

- Karagoz, O. K., Kilic, A., **Aydin, E. Y.,** Ankarali, M. M., & Uyanik, I. (2024). Dataset and Code for the Article: Predictive Uncertainty in State-Estimation Drives Active Sensing [Dataset and code]. Aperta. doi:10.48623/aperta.274062
- Aydin, E. Y.,** Unlutabak, B., & Uyanik, I. (2025). Dataset for the Article: Flow Impairs Multisensory Tracking and Increases Active Sensing in Weakly Electric Fish [Dataset]. Aperta. doi:10.48623/aperta.286374

Conferences and Scientific Meetings Attended

- 34th Signal Processing and Communications Applications Conference (SIU 2026), İstanbul, Türkiye, 7–10 July 2026
- SOAR/BISCCITs 2026, Wageningen, Netherlands, 22–24 June 2026
- I. International Neuronal Energy Dynamics Congress, Ankara, Türkiye, 26–28 November 2025
- Neuroscience 2025 (SFN 2025), San Diego, CA, USA, 15–19 November 2025
- 23rd National Neuroscience Congress (USK 2025), İzmir, Türkiye, 28–31 May 2025
- IFAC Conference on Cyber-Physical and Human Systems (CPHS 2024), Antalya, Türkiye, 12–13 December 2024
- Neuroscience 2024 (SFN 2024), Chicago, IL, USA, 5–9 October 2024
- FENS Forum 2024, Vienna, Austria, 25–29 June 2024
- 32nd Signal Processing and Communications Applications Conference (SIU 2024), Tarsus, Mersin, Türkiye, 15–18 May 2024
- Annual Meeting of SICB 2024, Seattle, WA, USA, 2–6 January 2024
- Neuroscience 2023 (SFN 2023), Washington, DC, USA, 11–15 November 2023
- 31st Signal Processing and Communications Applications Conference (SIU 2023), İstanbul, Türkiye, 5–8 July 2023
- FENS–Chen Institute–NeuroLéman Summer School 2023, Lausanne, Switzerland, 2–8 July 2023
- Annual Meeting of SICB 2023, Austin, TX, USA, 3–7 January 2023
- 20th National Neuroscience Congress (USK 2022), İstanbul, Türkiye, 19–23 October 2022
- FENS Forum 2022, Paris, France, 9–13 July 2022
- 30th Signal Processing and Communications Applications Conference (SIU 2022), Safranbolu, Karabük, Türkiye, 15–18 May 2022

Professional Development and International Training

SOAR/BISCCITs Workshop 2026

Wageningen, Netherlands

FUNDED PARTICIPANT; LIGHTNING-TALK AND POSTER PRESENTER

22–24 Jun 2026

- Participated in keynote lectures, tutorials, poster sessions, collaborative discussions, and laboratory tours focused on biologically inspired sensing and control. Presented doctoral research on active sensing in weakly electric fish through both a lightning talk and a scientific poster.

METEOR – Methodologies for Teamworking in Eco-Outwards Research

Online / European consortium

PARTICIPANT IN AN EU-FUNDED ONLINE PROFESSIONAL-DEVELOPMENT AND COLLABORATIVE RESEARCH PROGRAMME

2026 – Present

- Work in an international interdisciplinary group through regular virtual meetings and collaborative project development. The programme develops transferable competencies in teamwork, research planning, communication, and preparation of a team research and innovation project.

FENS–Chen Institute–NeuroLéman Summer School

Lausanne, Switzerland

SELECTED PARTICIPANT; FULL-STIPEND RECIPIENT

2–8 Jul 2023

- Completed an intensive international research school on the neural control of movement and sensory systems. Participated in lectures, scientific discussions, networking activities, and laboratory visits at EPFL, and applied the resulting perspectives to doctoral research on sensorimotor control.

Laboratory Animal Use in Experimental Research

Istanbul Medipol University,

Istanbul, Türkiye

CERTIFIED 80-HOUR COURSE

2019

- Completed the theoretical and practical certification required in Türkiye for conducting research involving laboratory animals, covering research ethics, animal welfare, handling, and experimental procedures.

Erasmus+ Training Programmes in Musculoskeletal Regenerative Research

Germany, Türkiye, Netherlands,

and Italy

INTERNATIONAL PARTICIPANT

2017–2019

- **Biomechanics Workshop**, Heidelberg University, Germany (1–5 Jul 2019): examined biomechanical approaches relevant to human movement and musculoskeletal research.
- **Emerging Nanotechnology Approaches in Musculoskeletal Tissue Engineering and Regenerative Medicine**, Ankara Yıldırım Beyazıt University, Türkiye (26–30 Nov 2018): studied nanotechnology-based strategies for tissue engineering and regenerative medicine.
- **Bench to Bedside and Back**, Utrecht University, Netherlands (16–20 Apr 2018): participated in a translational-research programme connecting laboratory findings, preclinical studies, and clinical applications.
- **Translational Postgraduate Education and Training Programme in Musculoskeletal Regenerative Research**, University of Bologna, Italy (6–10 Nov 2017): received international postgraduate training in translational musculoskeletal research.

Awards, Fellowships, & Grants

- **IBRO/SfN Travel Grant for Neuroscience 2026** – International Brain Research Organization and Society for Neuroscience; USD 2,500 (2026).
- **TÜBİTAK 2211-A National PhD Scholarship Program** – The Scientific and Technological Research Council of Türkiye; monthly scholarship (2021–2026).
- **Full Stipend, FENS–Chen Institute–NeuroLéman Summer School** – Federation of European Neuroscience Societies; EUR 540 (2023).
- **Full Stipend, SOAR/BISCCITs Workshop** – SOAR/BISCCITs Workshop Committee; EUR 1,500 (2026).

Professional Memberships: Federation of European Neuroscience Societies; Neuroscience Society of Turkey.